



From Accounts to Alerts: EDA and Logistic Regression for Early Corporate Risk Prediction

Yun-Hsuan Chang, Chia-Yen Chang, Huei-Wen Teng



Outline

1. Motivation
2. Data
3. Methodology



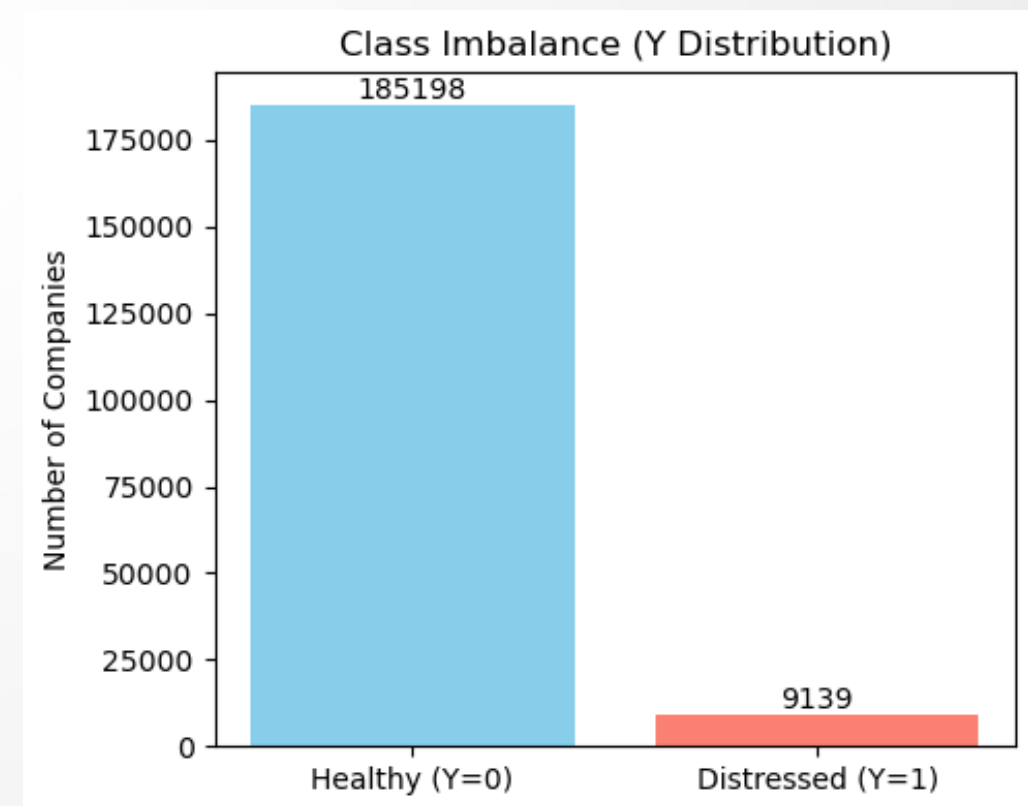
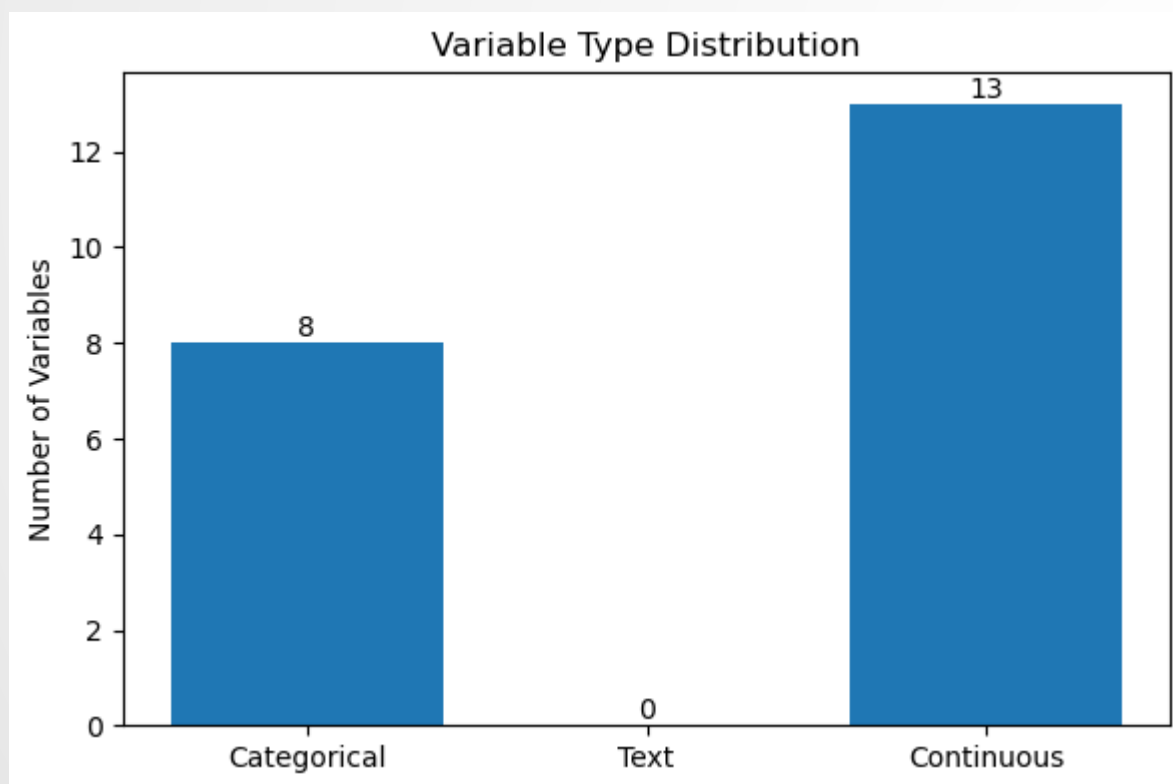
Motivation

- Early Warning Systems
- Transparency and Interpretability
- Publicly Available Data



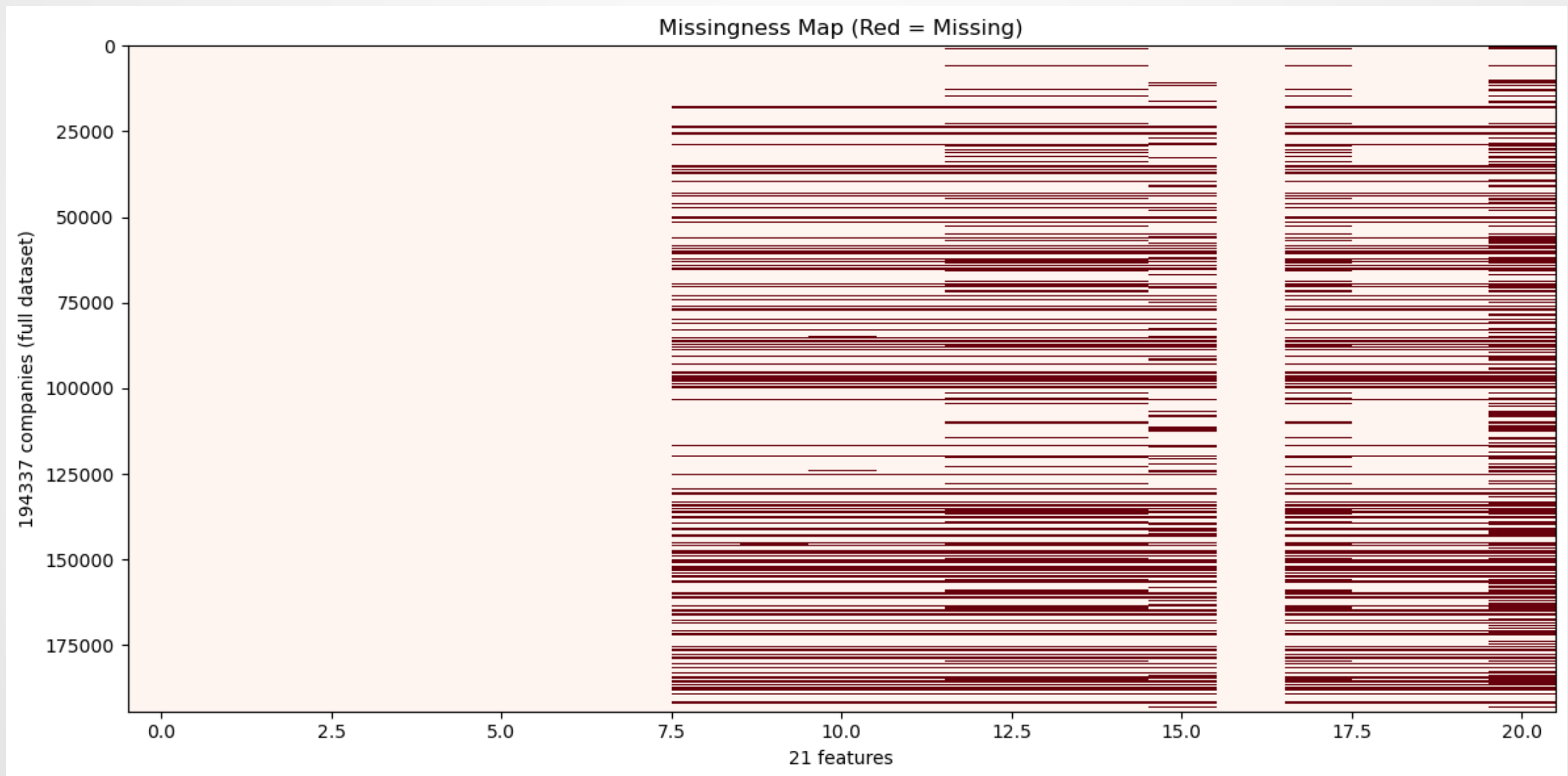
Data

- companies: 23033
- features: 21
- Class imbalance (Y=1 ratio): 0.0470



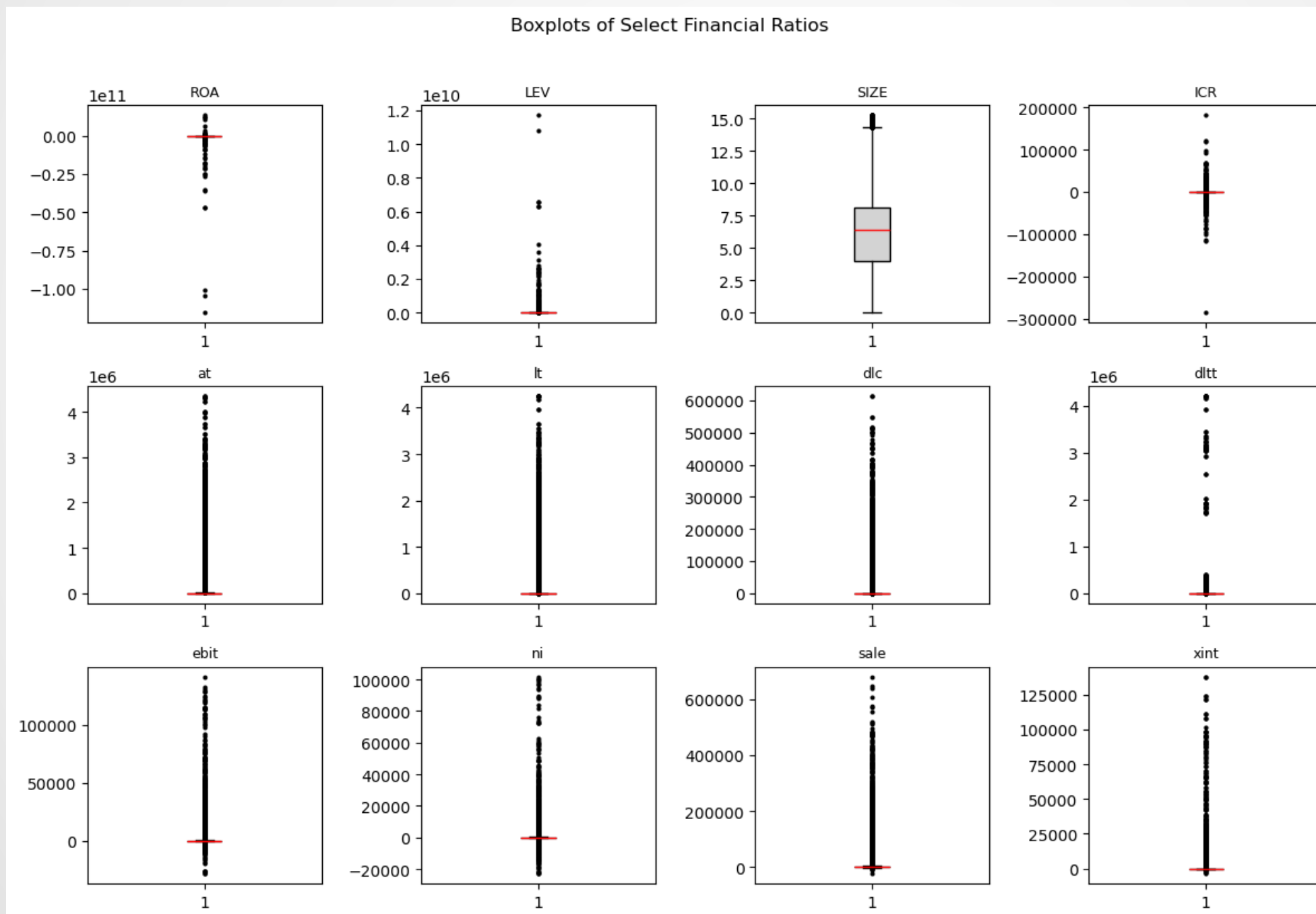
Data

- Average missing rate across all features: 18.17%
- Impute: SIC2×year median (train-only); else overall train median



Data

- heavy-tailed and asymmetric distributions



Data

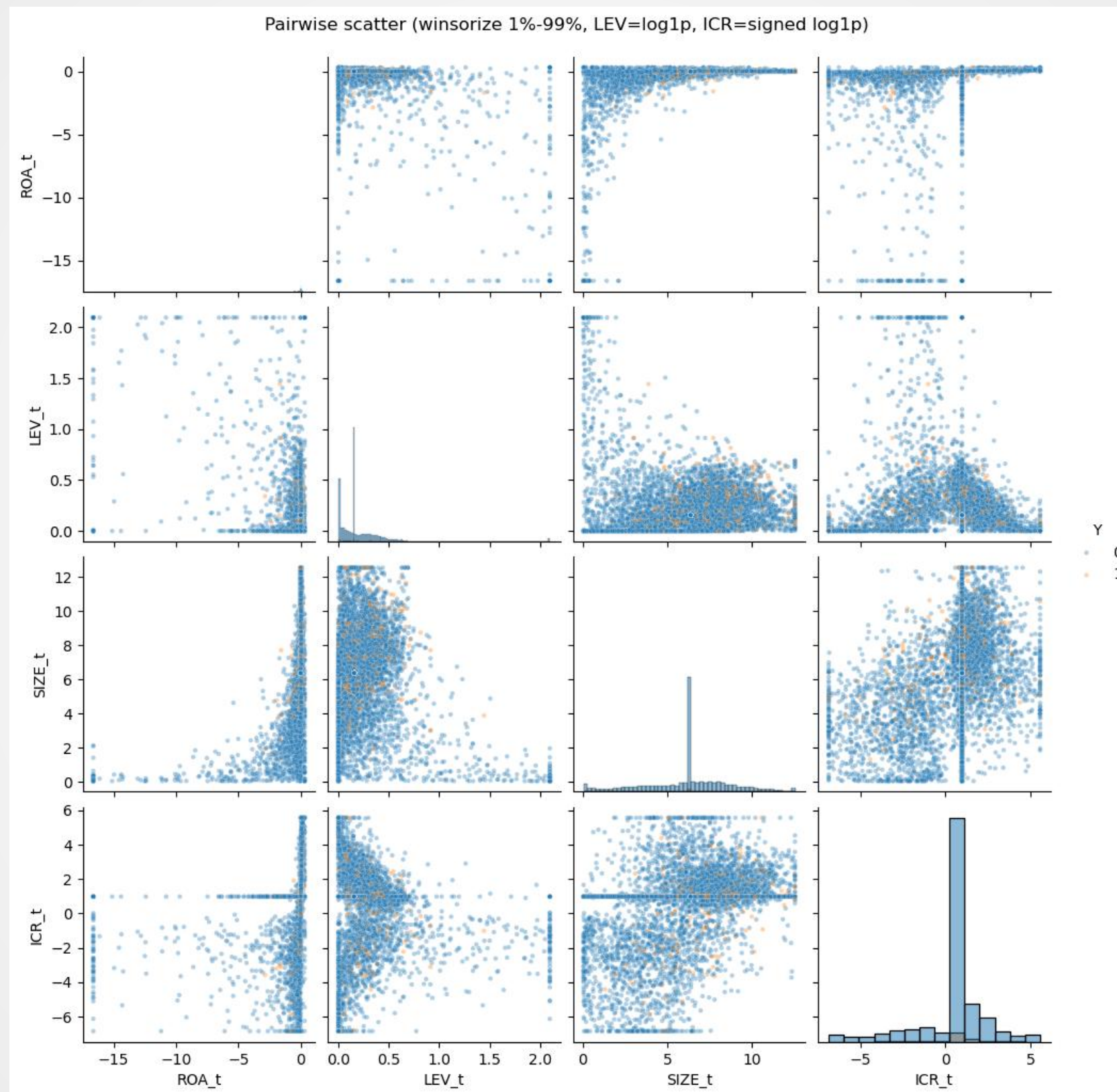
- Most variables are highly skewed
- Several variables exhibit extreme kurtosis (>10)
- apply winsorization and log-transform before model estimation

[Descriptive Summary of Selected Numerical Features]

	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
ROA	126632.0	-6869530.06	6.125031e+08	-1.154960e+11	-0.19	0.00	0.05	1.350200e+10	-140.97	23346.64
LEV	143774.0	1144596.71	6.431927e+07	0.000000e+00	0.03	0.17	0.41	1.172300e+10	113.36	16453.89
SIZE	143956.0	6.12	2.960000e+00	0.000000e+00	4.02	6.37	8.14	1.529000e+01	-0.08	-0.35
ICR	94503.0	-54.03	2.173370e+03	-2.855470e+05	-4.85	1.67	6.85	1.817916e+05	-19.70	4470.59
at	143956.0	20730.17	1.526165e+05	0.000000e+00	54.88	582.78	3439.39	4.349731e+06	14.35	247.71
lt	143773.0	17758.46	1.431946e+05	-1.000000e-01	17.28	315.03	2244.00	4.255074e+06	14.88	266.24
dlc	142956.0	1433.99	1.560927e+04	-5.600000e+01	0.00	4.41	62.90	6.142374e+05	19.34	449.17
dltt	143605.0	3902.35	6.283327e+04	-2.000000e-02	0.04	34.15	669.61	4.216909e+06	49.55	2719.41
ebit	126630.0	528.02	3.263470e+03	-2.838700e+04	-4.10	6.86	143.73	1.412900e+05	18.27	499.06
ni	126636.0	271.61	1.971740e+03	-2.311900e+04	-9.21	1.17	68.64	1.018320e+05	20.72	746.89
sale	126639.0	3703.27	1.790896e+04	-2.495468e+04	7.12	141.59	1219.69	6.809850e+05	13.86	283.16
xint	127418.0	211.92	2.356540e+03	-3.775000e+03	0.16	4.70	51.11	1.378610e+05	31.93	1255.55

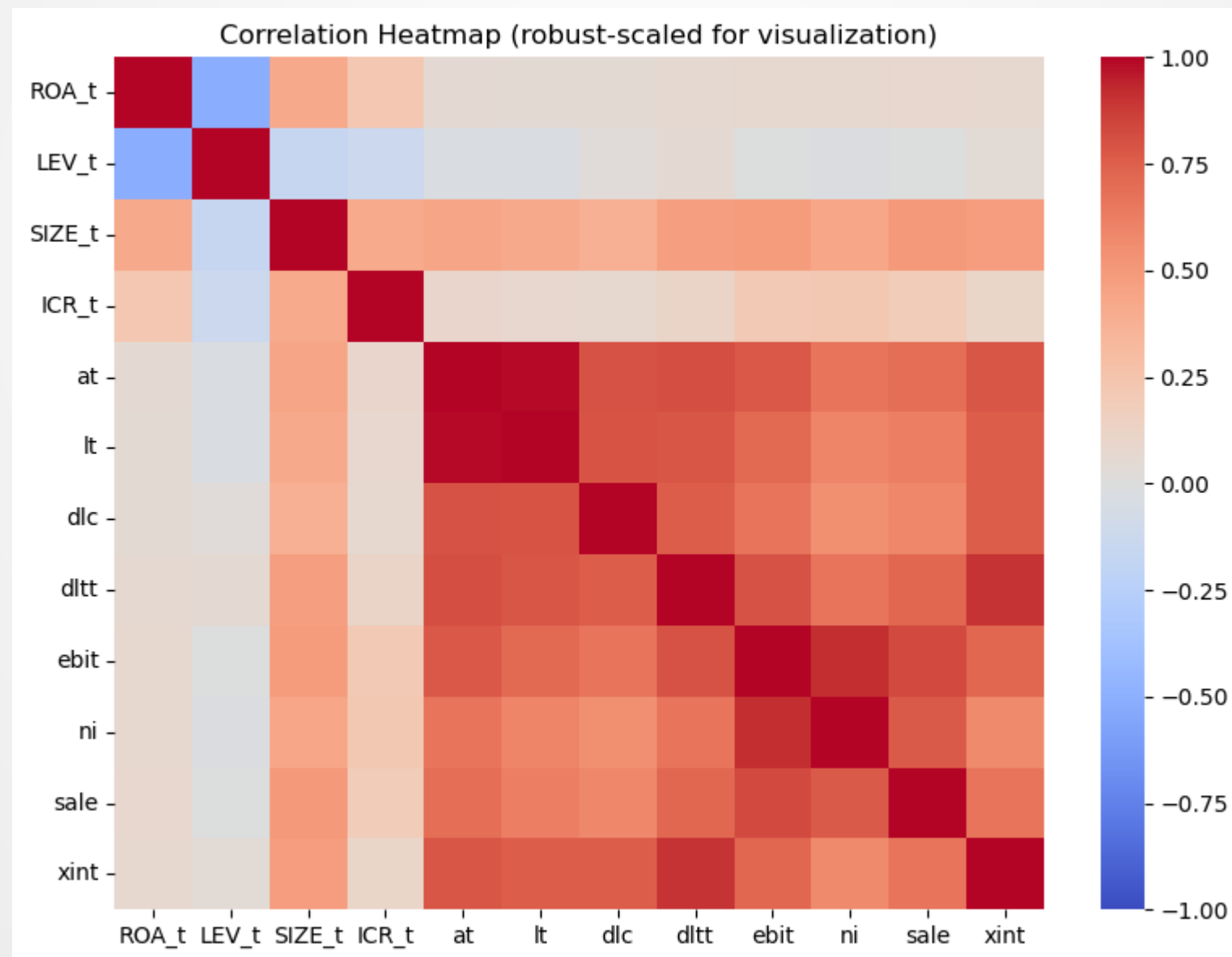


Data



Data

- Positive correlations among size-related variables (*at*, *lt*, *sale*, *ni*)
- ROA inversely related to LEV (higher leverage, lower profitability)
- No major multicollinearity after data transformation



Overview

- USA annual financial statements (2019–2024) + SEC Form 25 delisting labels
- Label: $Y=1$ if a Form 25 becomes effective within 400 days after the financial statement date
- Train: 2019–2023, Test: 2024



Benchmark Model

- Penalized logistic regression (L1 / Lasso and L2 / Ridge variants)
- Penalization reduces overfitting and selects relevant predictors



Class Imbalance Handling

- ▣ Synthetic Minority Oversampling Technique (SMOTE) applied
- ▣ Cost-sensitive weighting adjusts loss for rare default events



Design of Research

- EDA and Feature Engineering: cleaning, transforming, visualizing ratios
- Model Estimation: penalized logistic regression stratified by sector and time
- Evaluation and Interpretation: performance metrics and SHAP analysis



Performance Measures

- Discrimination: AUC / ROC, H-measure, and Precision–Recall curves
- Calibration: Brier score and calibration plots
- Threshold-based metrics: accuracy, sensitivity, specificity, F1-score





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